

BAGRAM, AfghanistanWMO: **697284**Lat: **34.944N**Lon: **69.259E**Elev: **1492**StdP: **84.64**Time Zone: **4.50 (AFG)**Period: **02-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.0	-19.8	0.8	6.1	-17.1	1.0	5.7	6.4	5.8	5.4	4.7	1.8	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	36.9	16.1	35.8	15.6	34.7	15.5	20.8	28.8	19.8	28.4	18.7	27.9	5.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	15.7	25.6	17.0	14.5	25.8	15.1	12.9	25.1	67.6	28.9	63.8	28.0	59.7	27.5	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.0	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	1.9	4.2	11.2	16.2	21.2	26.1	28.7	27.3	23.2	17.3	9.8	4.2
	DBStd	9.69	3.37	4.07	3.76	3.18	3.40	2.92	2.03	2.16	2.38	3.52	2.93	3.46
	HDD10.0	667	251	166	31	2	0	0	0	0	0	0	38	180
	HDD18.3	1972	509	396	221	77	12	0	0	0	1	64	255	437
	CDD10.0	2862	0	3	68	188	348	482	579	538	395	227	33	1
	CDD18.3	1126	0	0	1	13	102	232	321	279	146	33	0	0
	CDH23.3	14567	0	0	17	196	1114	2863	4504	3696	1759	413	4	0
CDH26.7	7029	0	0	1	41	394	1367	2499	1919	691	117	0	0	
Wind	WSAvg	2.9	1.4	1.5	2.1	2.9	3.9	5.2	5.2	4.1	3.0	2.3	1.4	1.5
Precipitation	PrecAvg	388	43	70	67	51	32	16	16	22	16	14	20	21
	PrecMax	657	102	139	167	122	130	58	47	88	47	58	76	70
	PrecMin	124	5	5	5	8	3	2	2	1	0	0	0	2
	PrecStd	121	28	36	40	30	27	14	10	18	14	17	19	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	16.9	25.1	30.2	34.0	37.3	38.4	37.8	34.2	32.0	24.0	16.9
		MCWB	4.6	7.9	11.0	12.8	14.1	16.0	16.9	16.3	15.3	12.4	10.2	6.5
	2%	DB	12.1	14.4	23.4	27.6	32.2	35.8	37.0	36.2	33.0	29.8	21.1	14.2
		MCWB	3.6	7.5	10.3	12.6	13.9	15.3	16.1	15.9	15.0	13.0	8.6	4.7
	5%	DB	10.2	12.8	21.2	25.8	30.9	34.2	36.0	35.1	31.9	27.9	18.9	13.0
		MCWB	2.6	6.7	9.4	12.1	13.7	14.9	15.7	15.8	14.6	12.4	7.8	4.0
	10%	DB	8.2	10.8	18.9	23.7	28.9	32.8	34.9	33.9	30.7	25.3	16.9	11.3
		MCWB	2.1	5.8	8.8	11.3	13.1	14.4	15.6	15.6	14.5	11.3	7.2	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	9.9	12.1	14.9	16.1	19.9	22.4	21.5	20.2	17.2	11.6	7.8
		MCDB	11.7	13.7	20.9	24.6	26.9	29.8	30.3	29.2	29.4	26.7	18.5	12.6
	2%	WB	4.6	8.6	11.4	13.8	15.3	16.8	21.1	20.8	18.7	15.2	10.5	6.7
		MCDB	9.3	12.3	19.5	24.0	26.0	32.1	29.0	28.5	27.3	23.8	17.2	11.1
	5%	WB	3.7	7.7	10.6	13.0	14.7	15.6	19.3	20.1	18.0	13.7	9.6	5.7
		MCDB	7.7	11.2	18.3	22.0	25.6	31.6	28.6	28.3	26.0	23.1	16.1	11.1
	10%	WB	2.9	6.8	9.9	12.4	14.1	15.0	17.5	19.4	17.1	12.4	8.6	4.2
		MCDB	6.8	10.2	17.1	20.7	25.5	31.0	30.5	27.9	25.4	22.8	14.2	9.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	8.7	11.2	12.3	12.3	12.2	12.6	13.1	14.4	14.8	13.1	12.3
		MCDBR	13.9	11.5	15.4	15.7	14.7	14.0	14.0	14.7	15.8	16.6	15.6	14.4
		MCWBR	7.2	5.5	5.7	4.9	4.7	5.0	5.7	5.8	6.1	6.4	6.5	7.2
	5% WB	MCDBR	10.7	9.3	12.3	12.9	12.9	13.6	13.0	12.1	12.4	14.3	12.5	12.0
		MCWBR	5.9	4.6	5.0	4.4	4.6	5.1	6.4	5.8	5.8	6.2	5.6	6.1
Clear-Sky Solar Irradiance	taub	0.310	0.360	0.422	0.456	0.462	0.474	0.475	0.442	0.390	0.376	0.341	0.312	
	taud	2.274	2.090	1.982	1.958	1.977	1.948	1.959	2.062	2.187	2.168	2.251	2.306	
	Ebn at Noon	884	868	841	832	831	818	814	835	865	841	837	858	
	Edh at Noon	103	139	169	182	181	187	183	163	136	127	105	94	
All-Sky Solar Radiation	RadAvg	2.86	3.59	4.75	5.81	7.05	7.95	8.01	7.35	6.38	4.92	3.29	2.71	
	RadStd	0.26	0.38	0.40	0.45	0.51	0.44	0.25	0.36	0.23	0.28	0.30	0.32	